INCH-POUND

MIL-C-39003/30
31 January 1992
SUPERSEDING
MIL-C-39003/3C
2 February 1984

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG), ESTABLISHED RELIABILITY, STYLE CSR23

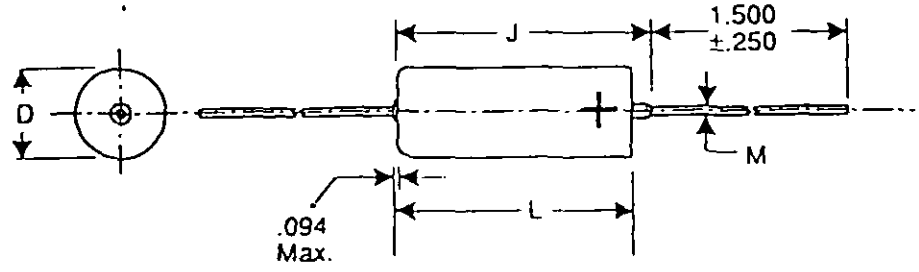
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-39003.

(D)

FAILURE RATE LEVELS "M", "P" AND "R" ARE INACTIVE FOR NEW DESIGN AFTER 2 FEBRUARY 1984.

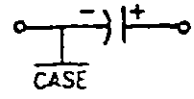
FAILURE RATE LEVEL "S" IS INACTIVE FOR NEW DESIGN AFTER 15 AUGUST 1984.



(D)

Case size	Dimensions L ±.031	D +.016 -.015	M ±.002	J max
A	.286	.135	.020	.422
B	.474	.185	.020	.610
C	.686	.289	.025	.822
D	.786	.351	.025	.922

Inches	mm	Inches	mm	Inches	mm
.002	.05	.135	3.43	.474	12.04
.015	.38	.185	4.70	.610	15.49
.016	.41	.250	6.35	.686	17.42
.020	.51	.286	7.26	.786	19.96
.025	.64	.289	7.34	.822	20.88
.031	.79	.351	8.92	.922	23.42
.094	2.39	.422	10.72	1.500	38.10



CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The case insulation shall extend .015 (0.38 mm) minimum beyond each end. However, when a shrink-fitted insulation is used, it shall lap over the ends of the capacitor body.
4. Lead length may be a minimum of 1-inch long for use in tape and reel automatic insertion equipment, when specified.

FIGURE 1. Dimensions and configurations.

(D) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:04Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Tubular, insulated.

Material - Metal (alloy unspecified).

Type of seal - Hermetic.

Leads: Types N-4, N-5, W-4, or W-5 of MIL-STD-1276.

DC rated voltage: See table I. Above +85°C voltage derating is required (see MIL-C-39003).

DC leakage (DCL): See table I.

Capacitance (Cap.)(nom): See table I.

Cap. tolerance: See table I.

Dissipation factor (DF): See table I.

Operating temperature: -55°C to +125°C.

Failure rate level: See table I.

Stability at low and high temperatures: In accordance with MIL-C-39003.

Step 1 (+25°C):

DCL - See table I.

Cap. - Within tolerance of table I value.

DF - See table I.

Step 2 (-55°C):

Cap. - Shall not change more than ±10 percent of step 1 measured value.

DF - See table I.

Step 3 (+25°C):

DCL - See table I.

Cap. - Shall not change more than ±2 percent of step 1 measured value.

DF - See table I.

Step 4 (+85°C):

DCL - See table I.

Cap. - Shall not change more than ±8 percent of step 1 measured value.

DF - See table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:04Z
Check the source to verify that this is the current version before use.

Step 5 (+125°C):

DCL - See table I.

Cap. - Shall not change more than ±12 percent of step 1 measured value.

DF - See table I.

Step 6 (+25°C):

DCL - See table I.

Cap. - Shall not change more than ±2 percent of step 1 measured value.

DF - See table I.

Surge voltage: In accordance with MIL-C-39003.

DCL - See table I.

Cap. - Shall not change more than ±2 percent of initial measured value.

DF - See table I.

Life: Method 108 of MIL-STD-202.

240, 1,000, and 2,000 hours:

At +25°C (as applicable):

DCL - See table I.

Cap. - Shall not change more than ±2 percent of initial measured value.

DF - See table I.

At +85°C:

DCL - See table I.

At +125°C:

DCL - See table I.

4,000, 6,000, 8,000, and 10,000 hours:

At +25°C (as applicable):

DCL - 200 percent of initial limit.

Cap. - Shall not change more than ±10 percent of initial measured value.

DF - See table I.

At +85°C:

DCL - 200 percent of initial limit.

At +125°C:

DCL - 200 percent of initial limit.

Marking: In accordance with MIL-C-39003.

Part or Identifying Number (PIN): M39003/03-(dash number from table I).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:04Z
Check the source to verify that this is the current version before use.

(D) TABLE I. Electrical characteristics and PIN's.

DC rated voltage	Capacitance (nom)	Capacitance tolerance	DC leakage at +25°C	DC leakage at +85°C	DC leakage at +125°C	Dissipation factor at -55°C +25°C	Dissipation factor at +25°C +85°C +125°C	Case size	PIN M39003/03- failure rate level for (%1,000 hrs.) M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
volts	µF	percent	µA	µA	µA	percent	percent								
6	10.0	10	0.9	9.0	11.0	6	6	A	0101	0201	0301	0401	2001	3001	4001
6	10.0	20	0.9	9.0	11.0	6	6	A	0102	0202	0302	0402	2002	3002	4002
6	12.0	10	1.0	10.0	12.5	6	6	A	0103	0203	0303	0403	2003	3003	4003
6	100.0	10	6.0	60.0	75.0	8	8	B	0104	0204	0304	0404	2004	3004	4004
6	100.0	20	6.0	60.0	75.0	8	8	B	0105	0205	0305	0405	2005	3005	4005
6	330.0	10	15.0	150.0	188.0	8	8	C	0106	0206	0306	0406	2006	3006	4006
6	330.0	20	15.0	150.0	188.0	8	8	C	0107	0207	0307	0407	2007	3007	4007
6	390.0	10	15.0	150.0	188.0	10	10	C	0108	0208	0308	0408	2008	3008	4008
6	470.0	10	15.0	150.0	188.0	10	10	C	0109	0209	0309	0409	2009	3009	4009
6	470.0	20	15.0	150.0	188.0	10	10	C	0110	0210	0310	0410	2010	3010	4010
6	680.0	10	20.0	200.0	250.0	10	10	D	0111	0211	0311	0411	2011	3011	4011
6	680.0	20	20.0	200.0	250.0	10	10	D	0112	0212	0312	0412	2012	3012	4012
6	820.0	10	20.0	200.0	250.0	10	10	D	0113	0213	0313	0413	2013	3013	4013
6	1000.0	10	30.0	300.0	375.0	10	10	D	0114	0214	0314	0414	2014	3014	4014
6	1000.0	20	30.0	300.0	375.0	10	10	D	0115	0215	0315	0415	2015	3015	4015
10	6.8	10	1.0	10.0	12.5	6	6	A	0116	0216	0316	0416	2016	3016	4016
10	6.8	20	1.0	10.0	12.5	6	6	A	0117	0217	0317	0417	2017	3017	4017
10	8.2	10	1.2	12.0	15.0	6	6	A	0118	0218	0318	0418	2018	3018	4018
10	47.0	10	5.0	50.0	63.0	6	6	B	0119	0219	0319	0419	2019	3019	4019
10	47.0	20	5.0	50.0	63.0	6	6	B	0120	0220	0320	0420	2020	3020	4020
10	56.0	10	6.0	60.0	75.0	6	6	B	0121	0221	0321	0421	2021	3021	4021
10	68.0	10	7.0	70.0	88.0	6	6	B	0122	0222	0322	0422	2022	3022	4022
10	68.0	20	7.0	70.0	88.0	6	6	B	0123	0223	0323	0423	2023	3023	4023
10	82.0	10	8.0	80.0	100.0	6	6	B	0124	0224	0324	0424	2024	3014	4014
10	220.0	10	15.0	150.0	188.0	8	8	C	0125	0225	0325	0425	2025	3015	4015
10	220.0	20	15.0	150.0	188.0	8	8	C	0126	0226	0326	0426	2026	3016	4016
10	270.0	10	15.0	150.0	188.0	8	8	C	0127	0227	0327	0427	2027	3017	4017
10	390.0	10	20.0	200.0	250.0	10	10	D	0128	0228	0328	0428	2028	3018	4018
10	470.0	10	20.0	200.0	250.0	10	10	D	0129	0229	0329	0429	2029	3029	4029
10	470.0	20	20.0	200.0	250.0	10	10	D	0130	0230	0330	0430	2030	3030	4030
10	560.0	10	30.0	300.0	375.0	10	10	D	0131	0231	0331	0431	2031	3031	4031

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T16:04Z
Check the source to verify that this is the current version before use.

TABLE I. Style CSR13 characteristics and PIN's - Continued.

DC rated voltage	Capacitance (nom)	Capacitance tolerance	DC leakage at +25°C	DC leakage at +85°C	DC leakage at +125°C	Dissipation factor at -55°C +25°C	Dissipation factor at +25°C +85°C +125°C	Case size	PIN M39003/03- failure rate level for (%1,000 hrs.) M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
volts	µF	percent	µA	µA	µA	percent	percent								
15	4.7	10	1.0	10.0	12.5	4	4	A	0132	0232	0332	0432	2032	3032	4032
15	4.7	20	1.0	10.0	12.5	4	4	A	0133	0233	0333	0433	2033	3033	4033
15	5.6	10	1.3	13.0	16.5	4	4	A	0134	0234	0334	0434	2034	3034	4034
15	33.0	10	6.0	60.0	75.0	6	6	B	0135	0235	0335	0435	2035	3035	4035
15	33.0	20	6.0	60.0	75.0	6	6	B	0136	0236	0336	0436	2036	3036	4036
15	39.0	10	6.0	60.0	75.0	6	6	B	0137	0237	0337	0437	2037	3037	4037
15	150.0	10	15.0	150.0	188.0	8	8	C	0138	0238	0338	0438	2038	3038	4038
15	150.0	20	15.0	150.0	188.0	8	8	C	0139	0239	0339	0439	2039	3039	4039
15	180.0	10	15.0	150.0	188.0	8	8	C	0140	0240	0340	0440	2040	3040	4040
15	220.0	10	20.0	200.0	250.0	8	8	D	0141	0241	0341	0441	2041	3041	4041
15	220.0	20	20.0	200.0	250.0	8	8	D	0142	0242	0342	0442	2042	3042	4042
15	270.0	10	20.0	200.0	250.0	8	8	D	0143	0243	0343	0443	2043	3043	4043
15	330.0	10	20.0	200.0	250.0	8	8	D	0144	0244	0344	0444	2044	3044	4044
15	330.0	20	20.0	200.0	250.0	8	8	D	0145	0245	0345	0445	2045	3045	4045
20	2.7	10	0.8	8.0	10.0	4	4	A	0146	0246	0346	0446	2046	3046	4046
20	3.3	10	1.0	10.0	12.5	4	4	A	0147	0247	0347	0447	2047	3047	4047
20	3.3	20	1.0	10.0	12.5	4	4	A	0148	0248	0348	0448	2048	3048	4048
20	3.9	10	1.2	12.0	15.0	4	4	A	0149	0249	0349	0449	2049	3049	4049
20	18.0	10	4.0	40.0	50.0	6	6	B	0150	0250	0350	0450	2050	3050	4050
20	22.0	10	4.0	40.0	50.0	6	6	B	0151	0251	0351	0451	2051	3051	4051
20	22.0	20	4.0	40.0	50.0	6	6	B	0152	0252	0352	0452	2052	3052	4052
20	27.0	10	5.0	50.0	63.0	6	6	B	0153	0253	0353	0453	2053	3053	4053
20	56.0	10	9.0	90.0	110.0	6	6	C	0154	0254	0354	0454	2054	3054	4054
20	68.0	10	10.0	100.0	125.0	6	6	C	0155	0255	0355	0455	2055	3055	4055
20	68.0	20	10.0	100.0	125.0	6	6	C	0156	0256	0356	0456	2056	3056	4056
20	82.0	10	10.0	100.0	125.0	6	6	C	0157	0257	0357	0457	2057	3057	4057
20	100.0	10	15.0	150.0	188.0	6	6	C	0158	0258	0358	0458	2058	3058	4058
20	100.0	20	15.0	150.0	188.0	6	6	C	0159	0259	0359	0459	2059	3059	4059
20	120.0	10	15.0	150.0	188.0	6	6	C	0160	0260	0360	0460	2060	3060	4060
20	150.0	10	20.0	200.0	250.0	8	8	D	0161	0261	0361	0461	2061	3061	4061
20	150.0	20	20.0	200.0	250.0	8	8	D	0162	0262	0362	0462	2062	3062	4062
20	180.0	10	20.0	200.0	250.0	8	8	D	0163	0263	0363	0463	2063	3063	4063

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:04Z
Check the source to verify that this is the current version before use.

TABLE I. Style CSR13 characteristics and PIN's - Continued.

DC rated voltage	Capacitance (nom)	Capacitance tolerance	DC leakage at +25°C	DC leakage at +85°C	DC leakage at +125°C	Dissipation factor at -55°C +25°C	Dissipation factor at +25°C +85°C +125°C	Case size	PIN M39003/03- failure rate level for (%1,000 hrs.) M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
volts	µF	percent	µA	µA	µA	percent	percent								
35	1.8	10	1.0	10.0	12.5	4	4	A	0164	0264	0364	0464	2064	3064	4064
35	8.2	10	3.5	35.0	44.0	6	6	B	0165	0265	0365	0465	2065	3065	4065
35	10.0	10	4.0	40.0	50.0	6	6	B	0166	0266	0366	0466	2066	3066	4066
35	10.0	20	4.0	40.0	50.0	6	6	B	0167	0267	0367	0467	2067	3067	4067
35	33.0	10	10.0	100.0	125.0	6	6	C	0168	0268	0368	0468	2068	3068	4068
35	33.0	20	10.0	100.0	125.0	6	6	C	0169	0269	0369	0469	2069	3069	4069
35	39.0	10	10.0	100.0	125.0	6	6	C	0170	0270	0370	0470	2070	3070	4070
35	47.0	10	10.0	100.0	125.0	6	6	C	0171	0271	0371	0471	2071	3071	4071
35	47.0	20	10.0	100.0	125.0	6	6	C	0172	0272	0372	0472	2072	3072	4072
35	56.0	10	15.0	150.0	188.0	6	6	D	0173	0273	0373	0473	2073	3073	4073
35	68.0	10	15.0	150.0	188.0	6	6	D	0174	0274	0374	0474	2074	3074	4074
35	68.0	20	15.0	150.0	188.0	6	6	D	0175	0275	0375	0475	2075	3075	4075
50	1.2	10	0.9	9.0	11.0	4	4	A	0176	0276	0376	0476	2076	3076	4076
50	1.5	10	1.2	12.0	15.0	4	4	A	0177	0277	0377	0477	2077	3077	4077
50	1.5	20	1.2	12.0	15.0	4	4	A	0178	0278	0378	0478	2078	3078	4078
50	5.6	10	4.5	45.0	56.0	4	4	B	0179	0279	0379	0479	2079	3079	4079
50	6.8	10	4.5	45.0	56.0	6	6	B	0180	0280	0380	0480	2080	3080	4080
50	6.8	20	4.5	45.0	56.0	6	6	B	0181	0281	0381	0481	2081	3081	4081
50	22.0	10	10.0	100.0	125.0	6	6	C	0182	0282	0382	0482	2082	3082	4082
50	22.0	20	10.0	100.0	125.0	6	6	C	0183	0283	0383	0483	2083	3083	4083
50	27.0	10	10.0	100.0	125.0	6	6	C	0184	0284	0384	0484	2084	3084	4084
50	33.0	10	10.0	100.0	125.0	6	6	D	0185	0285	0385	0485	2085	3085	4085
50	33.0	20	10.0	100.0	125.0	6	6	D	0186	0286	0386	0486	2086	3086	4086
50	39.0	10	10.0	100.0	125.0	6	6	D	0187	0287	0387	0487	2087	3087	4087

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T16:04Z
Check the source to verify that this is the current version before use.

Substitutability data:

Substitutability for failure rate levels shall be as follows: In accordance with MIL-C-39003.

Substitutability for capacitance tolerance shall be as follows:

Cap tolerance	Will replace cap tolerance
10 percent	20 percent

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AH
Navy - SH
Air Force - 85, 99
DLA - ES

User activities:
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5910-1755)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:04Z
Check the source to verify that this is the current version before use.